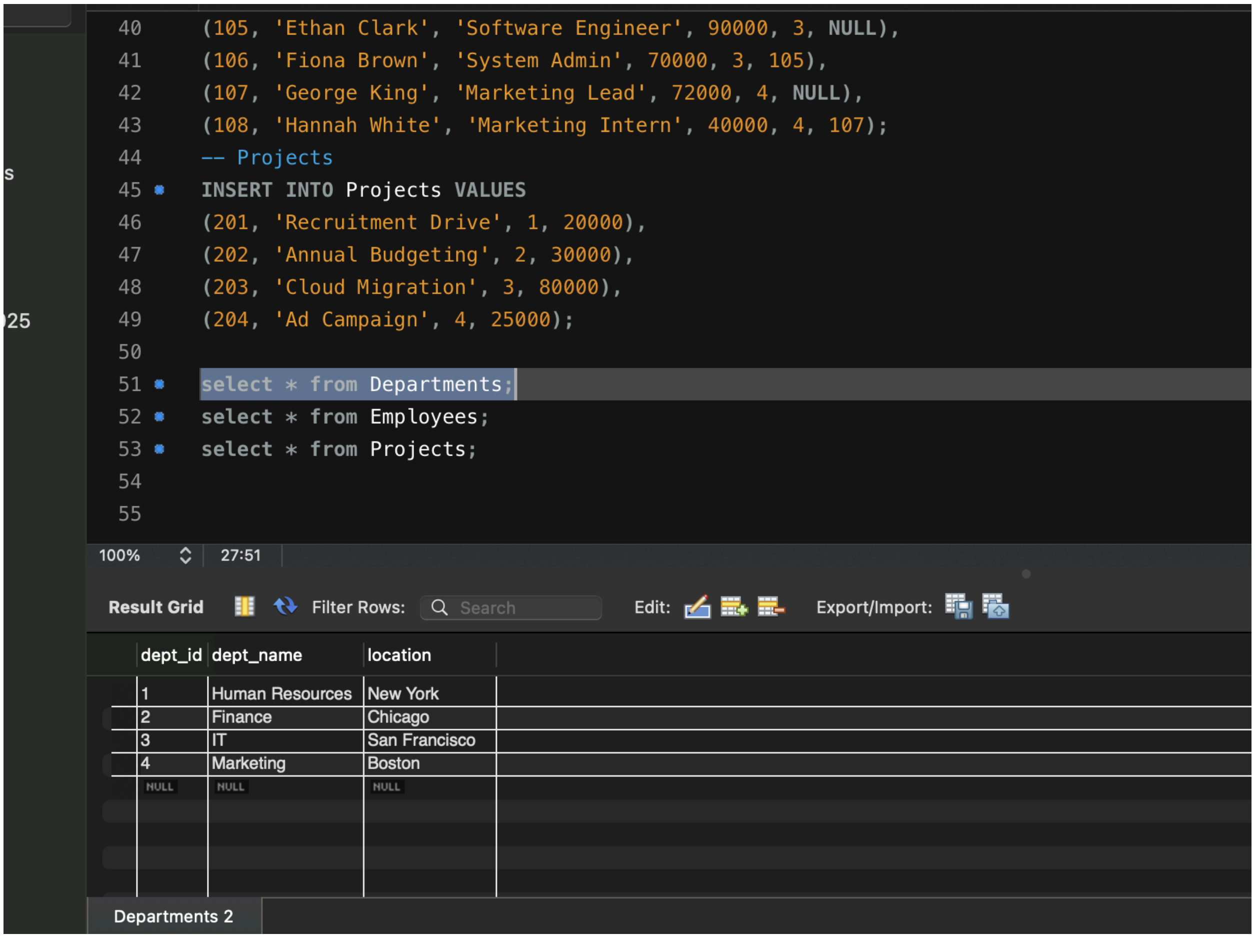
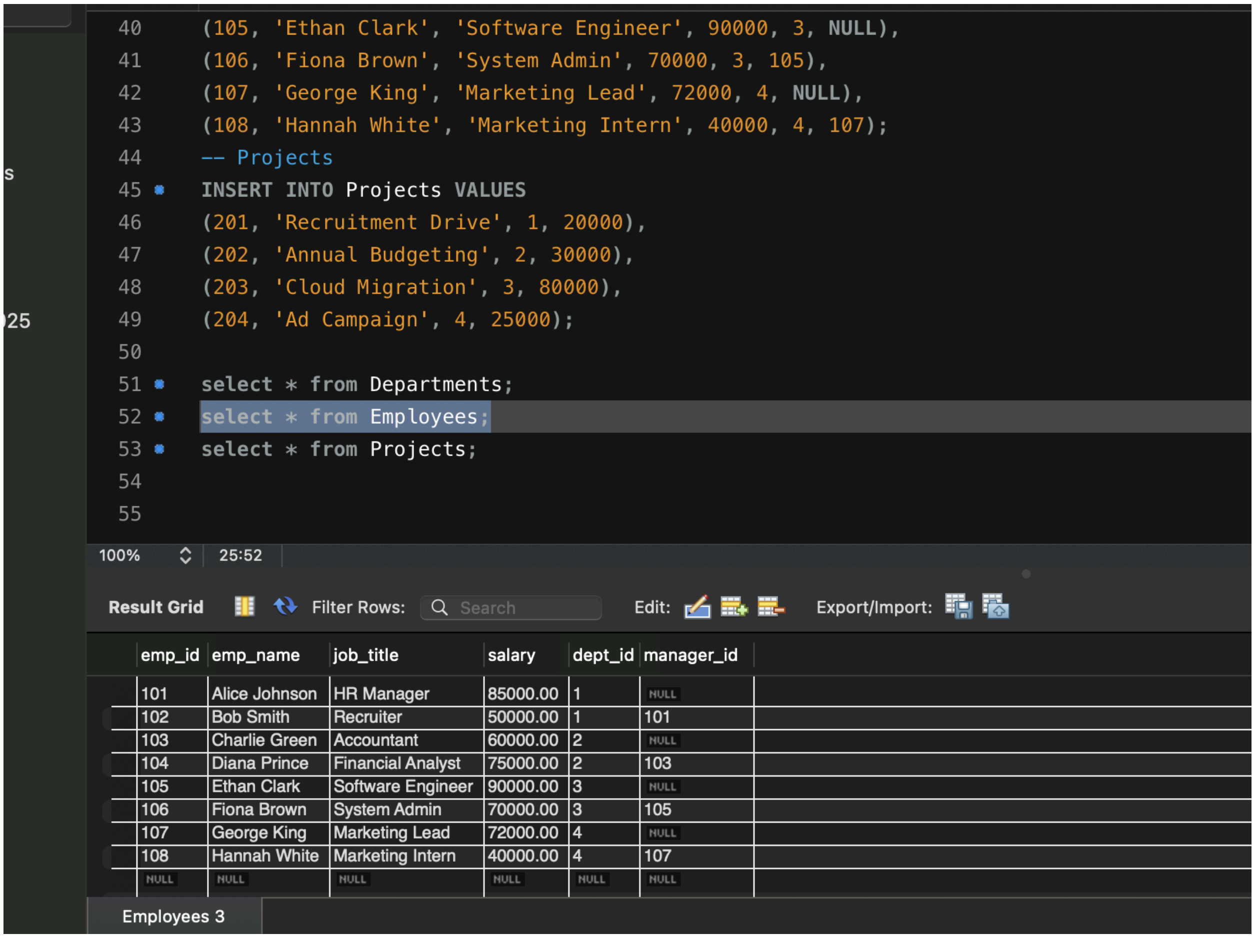
– Departments table



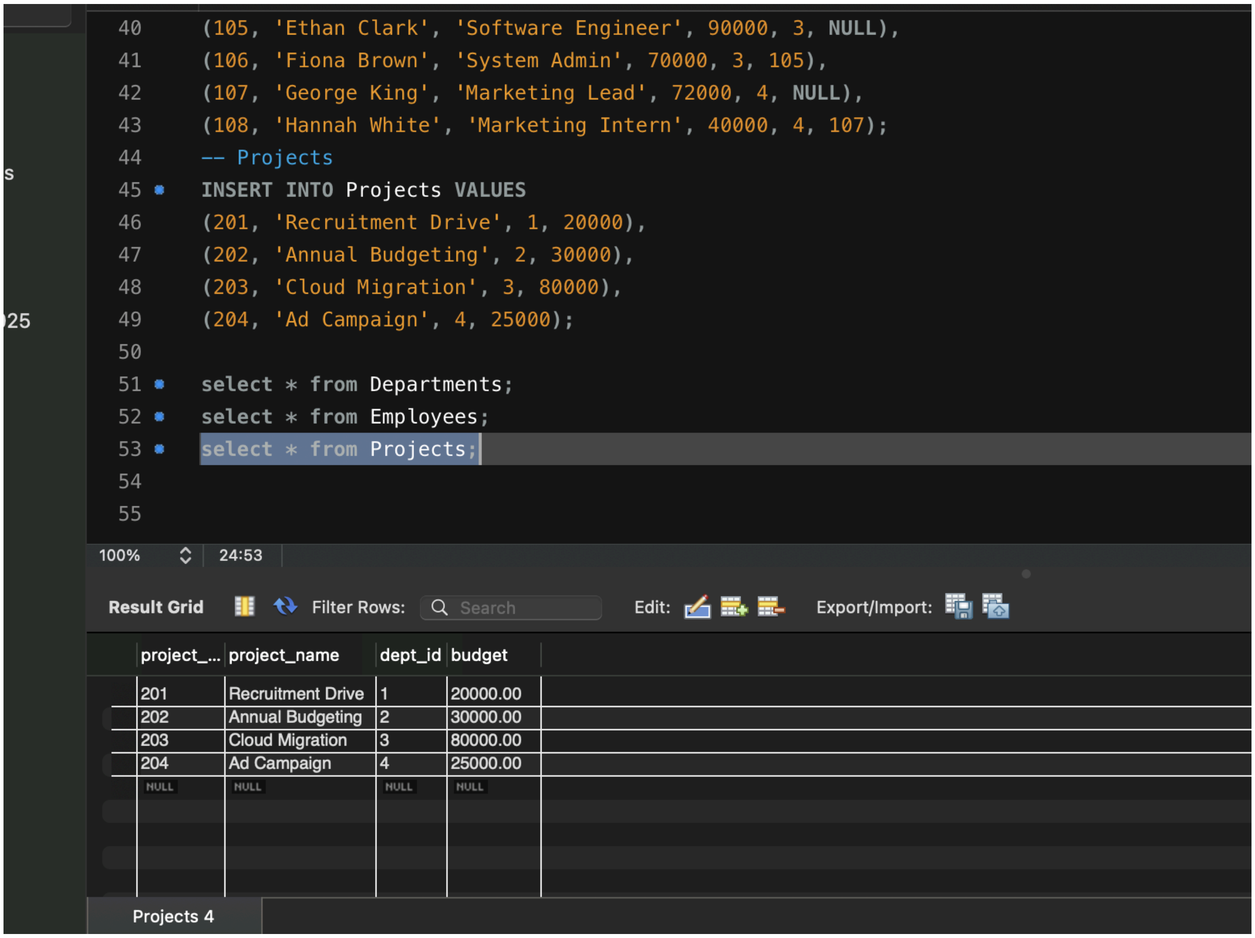
```sql
(105, 'Ethan Clark', 'Software Engineer', 90000, 3, NULL),
(106, 'Fiona Brown', 'System Admin', 70000, 3, 105),
(107, 'George King', 'Marketing Lead', 72000, 4, NULL),
(108, 'Hannah White', 'Marketing Intern', 40000, 4, 107);
-- Projects
INSERT INTO Projects VALUES
(201, 'Recruitment Drive', 1, 20000),
(202, 'Annual Budgeting', 2, 30000),
(203, 'Cloud Migration', 3, 80000),
(204, 'Ad Campaign', 4, 25000);

select * from Departments;
select * from Employees;
select * from Projects;
```


dept_id	dept_name	location
1	Human Resources	New York
2	Finance	Chicago
3	IT	San Francisco
4	Marketing	Boston
NULL	NULL	NULL

– Employees table

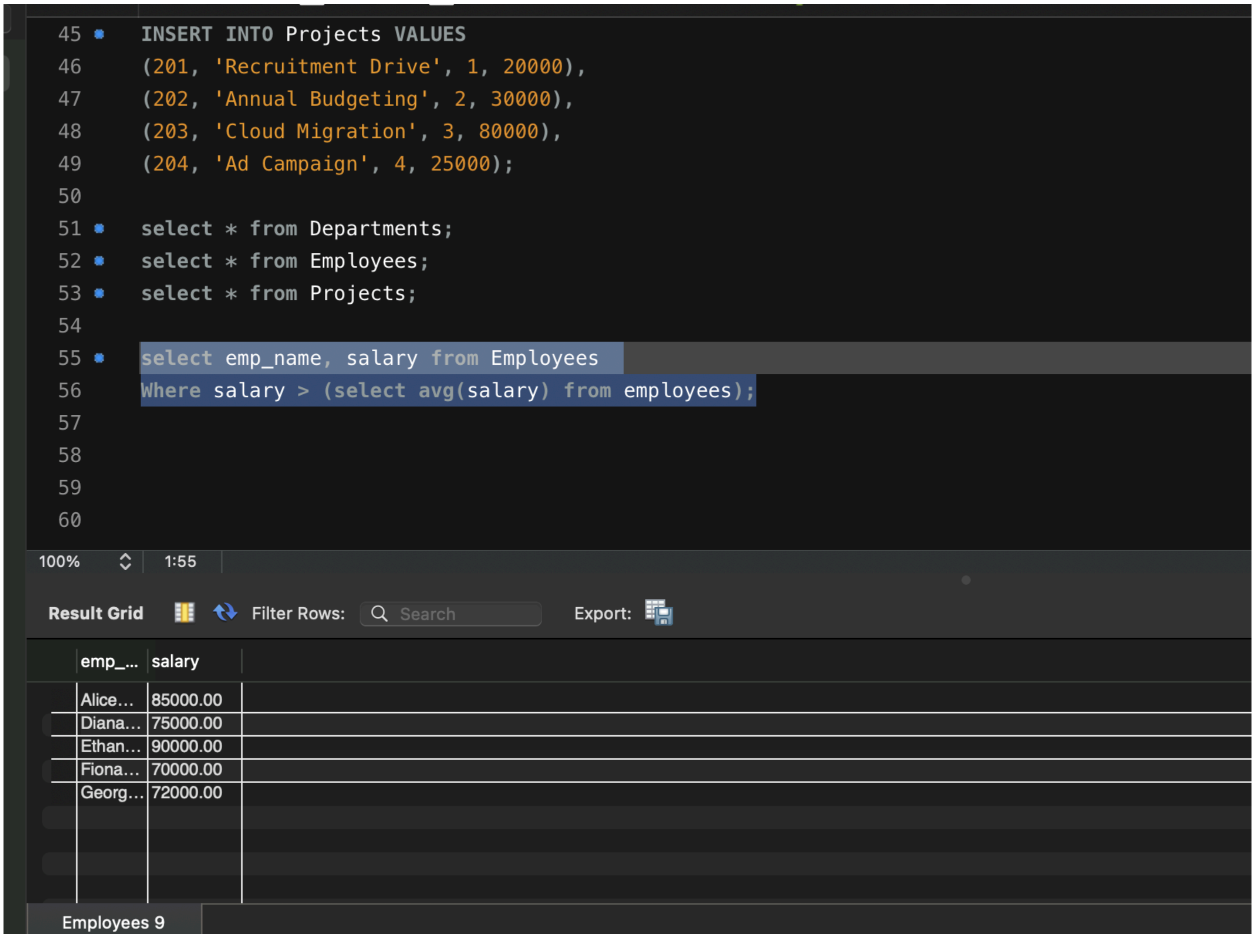


– Projects Table

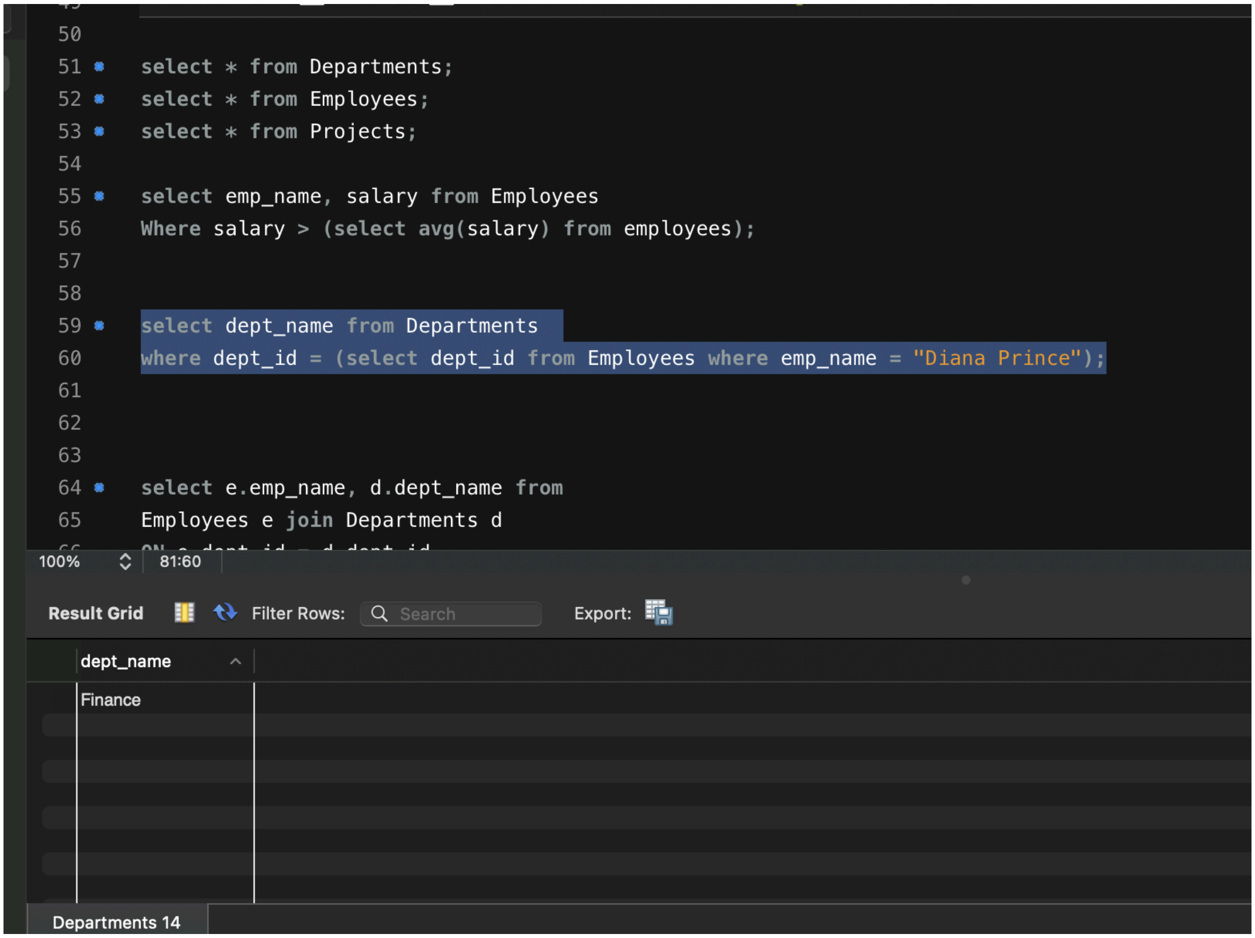


Q1.
Display the names of all employees who earn more than the average salary of all employees.

select emp_name, salary from Employees
Where salary > (select avg(salary) from employees);

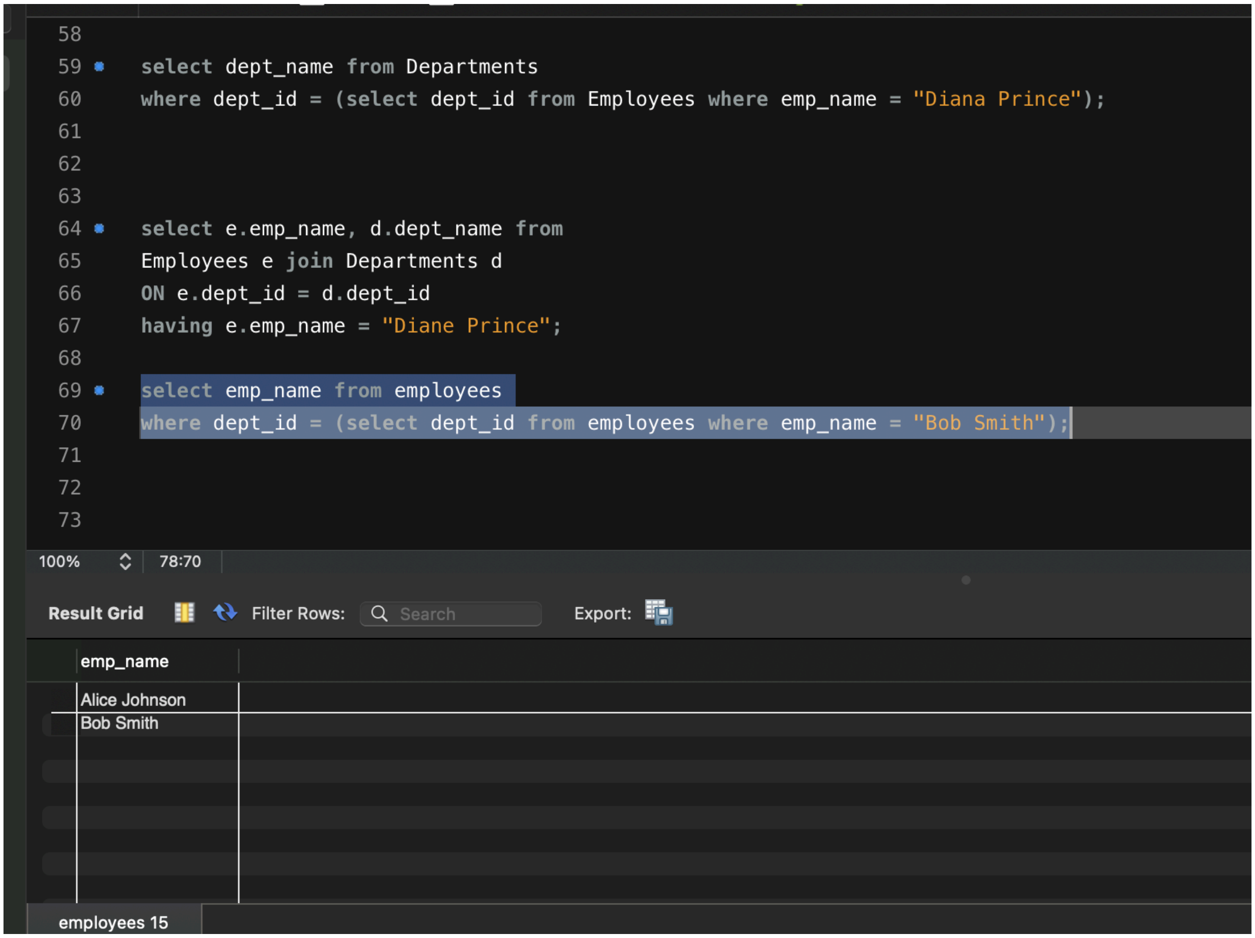


Q2. Find the department name of the employee 'Diana Prince'.
select dept_name from Departments
where dept_id = (select dept_id from Employees where emp_name = "Diana Prince");



List all employees who work in the
same department as
'Bob Smith'.

select emp_name from employees
where dept_id = (select dept_id from employees where emp_name = "Bob Smith");



Q4.
Find employees who have
a higher salary than
'Charlie Green'

```
select emp_name, salary from Employees
where salary > (select salary from Employees where emp_name = "Charlie Green");
```

```sql
where dept_id = (select dept_id from Employees where emp_name = "Diana Prince");



select e.emp_name, d.dept_name from
Employees e join Departments d
ON e.dept_id = d.dept_id
having e.emp_name = "Diane Prince";

select emp_name from employees
where dept_id = (select dept_id from employees where emp_name = "Bob Smith");

select emp_name, salary from Employees
where salary > (select salary from Employees where emp_name = "Charlie Green");
```

100% 80:73

Result Grid Filter Rows: Search Export:

emp_name	salary
Alice Johnson	85000.00
Diana Prince	75000.00
Ethan Clark	90000.00
Fiona Brown	70000.00
George King	72000.00

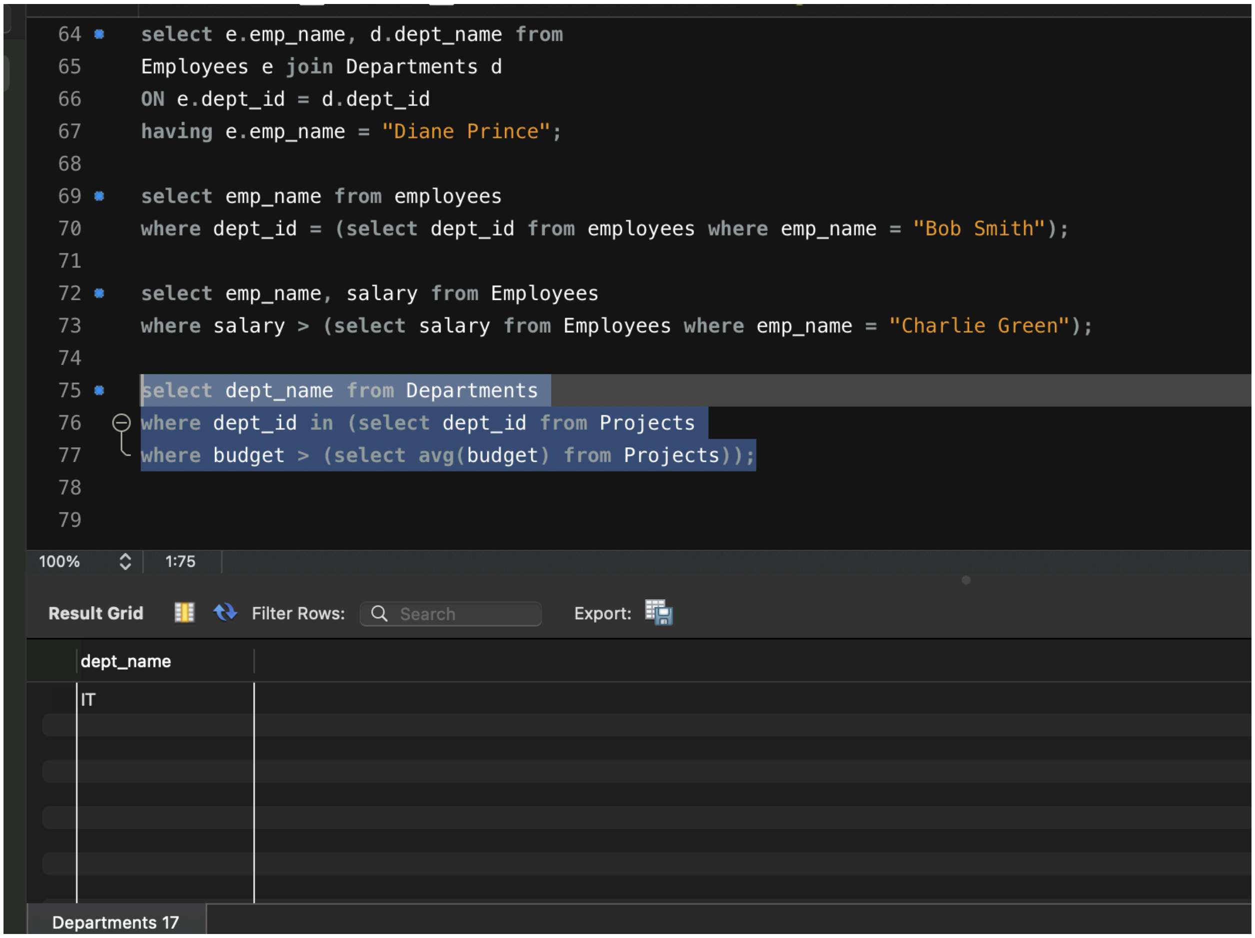
Employees 16

Q5.

Display the names of departments that have a project budget greater than the

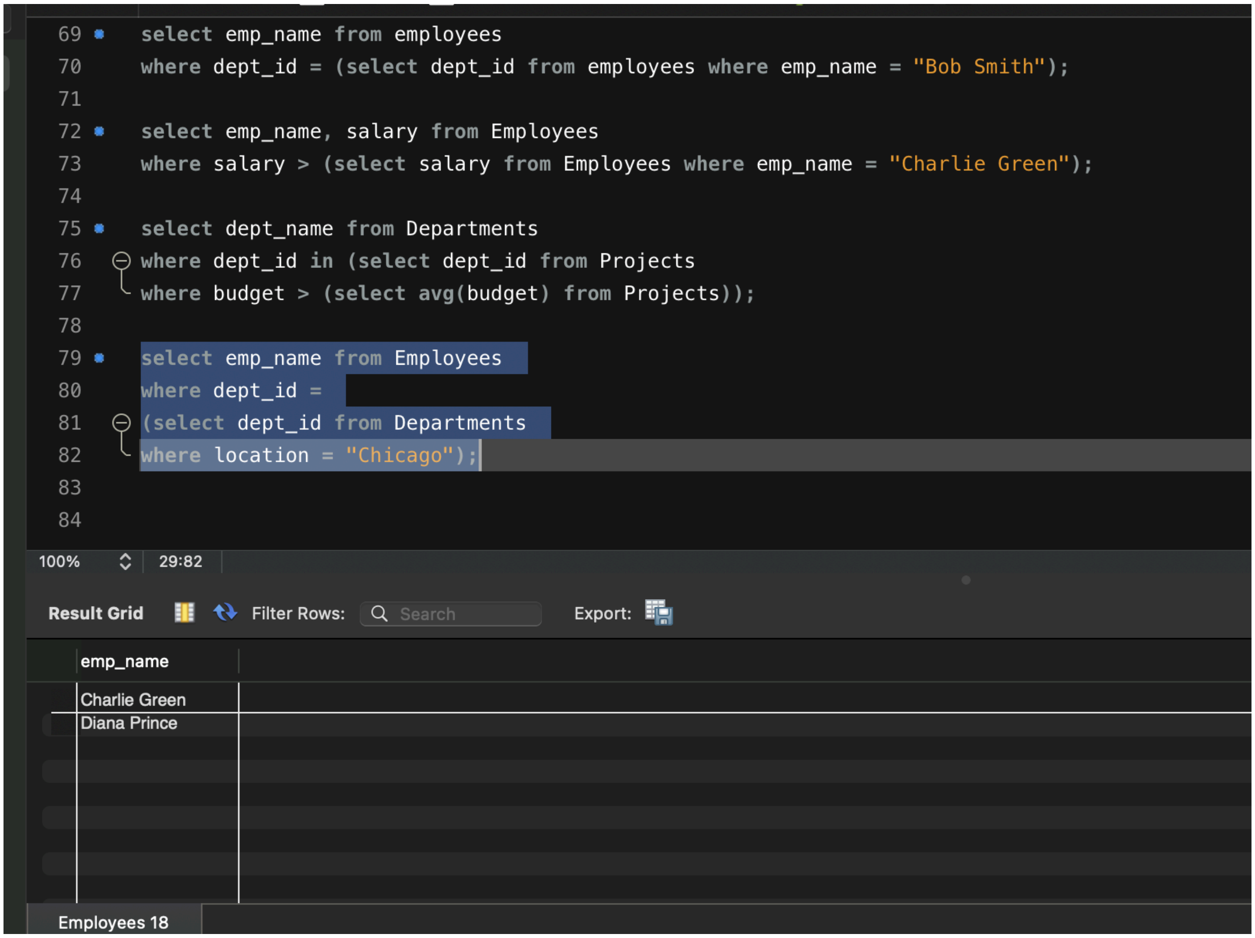
average project budget

select dept_name from Departments
where dept_id in (select dept_id from Projects
where budget > (select avg(budget) from Projects));



Q6.
List all employees whose
department is located in ‘Chicago’.

```
select emp_name from Employees
where dept_id =
(select dept_id from Departments
where location = "Chicago");
```



Q7.
Find the
highest-paid employee
in each department using a
correlated subquery

```
select emp_name, salary, dept_id from Employees
where (dept_id,salary) in (select dept_id, max(salary) from Employees
group by dept_id);
```

```sql
where salary > (select salary from Employees where emp_name = "Charlie Green");

select dept_name from Departments
where dept_id in (select dept_id from Projects
where budget > (select avg(budget) from Projects));

select emp_name from Employees
where dept_id =
(select dept_id from Departments
where location = "Chicago");

select emp_name, salary, dept_id from Employees
where (dept_id,salary) in (select dept_id, max(salary) from Employees
group by dept_id);
```

100% 1:84

Result Grid Filter Rows: Search Export:

emp_name	salary	dept_id
Alice Johnson	85000.00	1
Diana Prince	75000.00	2
Ethan Clark	90000.00	3
George King	72000.00	4

Employees 19